ScopePilot

REVIEW 2 PASSED

Date: Oct 15, 2026

Review 2 of 5 Technical Report: Rule-Based Mock & Structured Output

Student / Developer:	Vishal	Assigned Role:	AI / RAG / Change-Analysis Agent
Review Stage:	2 of 5 (Rule-Based Mock & Structured Output)	Git Repository:	github.com/Vivek-teja/ScopePilot
Working Branch:	`main`	Evaluation Status:	MOCK RUNNABLE • 5 CASES VERIFIED

1. Executive Summary & Objective

The primary goal of **Review 2** is to prove the end-to-end request/response contract with deterministic, reproducible behavior before introducing probabilistic LLMs or vector embeddings.

> **Review 2 Focus:** Construct a rule-based mock engine that accepts approved proposal scope items and deliverables alongside incoming client change requests. Exercise the five required case categories: (1) clearly in scope, (2) clearly out of scope, (3) partial overlap, (4) ambiguous request, and (5) insufficient evidence.

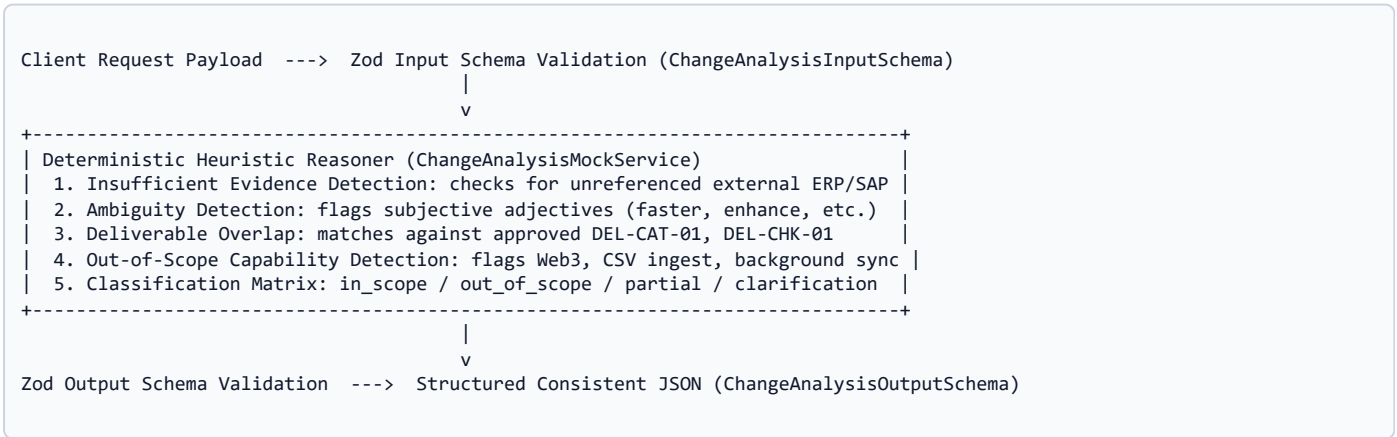
2. The Five Canonical Case Scenarios

Five canonical fixtures were developed under `backend/src/ai/fixtures/changeAnalysisFixtures.js` representing distinct evaluation outcomes:

Case ID	Category	Incoming Client Request	Expected Status	Confidence
Case 1	**Clearly In Scope**	Checkout phone number validation & E.164 formatting	`in_scope`	**95%** (High)
Case 2	**Clearly Out of Scope**	Cryptocurrency Wallet Integration & Web3 Token Settlement	`out_of_scope`	**92%** (High)
Case 3	**Partial Overlap**	Catalog CSV Bulk Import and Automated Inventory Sync	`partially_out_of_scope`	**88%** (High)
Case 4	**Ambiguous Request**	"Make store experience significantly faster and enhance look"	`needs_clarification`	**60%** (Moderate)
Case 5	**Insufficient Evidence**	Sync orders to internal enterprise ERP via custom connector	`uncertain`	**45%** (Low)

3. Architecture & Deterministic Mock Engine

The mock service (`backend/src/ai/services/changeAnalysisMock.service.js`) applies strict contract validation and deterministic heuristics:



> **Honesty & Safeguards:** For Ambiguous (Case 4) and Insufficient Evidence (Case 5) requests, the engine deliberately avoids forcing artificial certainty. It generates targeted clarifying questions for the client and returns lowered confidence scores (45%–60%).

4. CLI Demonstration Output (Summary Table)

Executing `npm run mock:cases` runs all five cases consecutively and checks determinism:

```
┌─────────┬──────────┬──────────────────────┬──────────────────────────────────────────────────┬──────────────────────┬────────────┬──────────┬──────────────┐
│ (index) │ caseId   │ category             │ requestTitle                                     │ reviewStatus         │ confidence │ inScope  │ outOfScope   │ uncertain │ deterministic │
├─────────┼──────────┼──────────────────────┼──────────────────────────────────────────────────┼──────────────────────┼────────────┼──────────┼──────────────┼───────────┼───────────────┤
│ 0       │ 'case-1' │ 'Clearly In Scope'   │ 'Checkout phone number validation...'            │ 'in_scope'           │ '95%'      │ 1        │ 0            │ 0         │ 'YES'         │
│ 1       │ 'case-2' │ 'Clearly Out of Scope' │ 'Cryptocurrency Wallet Integration & ...'      │ 'out_of_scope'       │ '92%'      │ 0        │ 7            │ 0         │ 'YES'         │
│ 2       │ 'case-3' │ 'Partial Overlap'    │ 'Product Catalog CSV Bulk Import and ...'        │ 'partially_out_of_scope' │ '88%'  │ 1        │ 4            │ 0         │ 'YES'         │
│ 3       │ 'case-4' │ 'Ambiguous Request'  │ 'Make store experience significantly ...'        │ 'needs_clarification'│ '60%'      │ 0        │ 0            │ 1         │ 'YES'         │
│ 4       │ 'case-5' │ 'Insufficient Evidence' │ 'Sync orders to internal enterprise E...'     │ 'uncertain'          │ '45%'      │ 0        │ 1            │ 6         │ 'YES'         │
└─────────┴──────────┴──────────────────────┴──────────────────────────────────────────────────┴──────────────────────┴────────────┴──────────┴──────────────┴───────────┴───────────────┘
```

5. Sample Structured JSON Outputs

A. Clearly In Scope (Case 1)

```
{
  "summary": "The requested change is fully aligned with existing deliverables in the approved proposal baseline.",
  "reviewStatus": "in_scope",
  "confidenceScore": 0.95,
  "inScopeItems": [{
    "item": "Checkout Flow and Payment Processing",
    "matchedDeliverableId": "del_chk_01",
    "matchedScopeReference": "DEL-CHK-01",
    "reason": "Request references functionality covered under approved deliverable \"Checkout Flow and Payment
Processing\"."
  }],
  "outOfScopeItems": [],
  "uncertainItems": [],
  "affectedDeliverables": [{ "deliverableId": "del_chk_01", "scopeReference": "DEL-CHK-01" }],
  "disclaimer": "AI analysis is an advisory draft only..."
}
```

B. Partial Overlap (Case 3)

```
{
  "summary": "The request touches approved deliverables (Product Catalog and Search UI) but introduces substantial new
out-of-scope features.",
  "reviewStatus": "partially_out_of_scope",
  "confidenceScore": 0.88,
  "inScopeItems": [{ "item": "Product Catalog and Search UI", "matchedScopeReference": "DEL-CAT-01" }],
  "outOfScopeItems": [
    { "item": "CSV Bulk Ingestion Engine", "reason": "Not included in approved proposal scope or deliverables." },
    { "item": "Automated Supplier Inventory Sync", "reason": "Requires background cron job and external adapter." }
  ],
  "affectedDeliverables": [{ "deliverableId": "del_cat_01", "scopeReference": "DEL-CAT-01" }]
}
```

C. Insufficient Evidence (Case 5 - Missing Documentation Flagged)


```
{
  "summary": "Insufficient evidence in the approved proposal to determine feasibility or baseline alignment for
requested external integrations.",
  "reviewStatus": "uncertain",
  "confidenceScore": 0.45,
  "uncertainItems": [{
    "item": "On-premise SAP ERP Integration",
    "question": "Can the client provide technical specifications, API schemas, and network requirements for the sap
integration?",
    "missingEvidence": "The approved proposal scope contains zero architecture documentation or interface
specifications for \"sap\"."
  }]
}
```


6. Implementation Deliverables & Code Changes

File Path	Description & Role in Review 2
`backend/src/ai/services/changeAnalysisMock.service.js`	Deterministic rule-based mock engine supporting all 5 case categories with reproducible outputs.
`backend/src/ai/fixtures/changeAnalysisFixtures.js`	Five canonical test cases with complete proposal snapshots, deliverables, and explanations.
`backend/src/ai/scripts/runMockCases.js`	CLI runner executable via `npm run mock:cases` with summary table and determinism verification.
`backend/src/ai/ai.controller.js`	Express controller exposing dynamic mock analysis and test case discovery/execution.
`backend/src/ai/ai.routes.js`	HTTP routes mounted under `/api/ai/*` with Zod request validation.
`backend/tests/ai.mockCases.test.js`	Dedicated Jest test suite testing all 5 canonical cases, constraints, and determinism.
`backend/tests/ai.routes.test.js`	Integration tests for `GET /api/ai/cases`, `POST /cases/:id/run`, and `POST /analyze-mock`.
`backend/src/ai/README.md`	Updated testing instructions, CLI documentation, and endpoint catalog.
`REVIEW_2_SUBMISSION.md`	Complete mentor review evaluation document.

7. Verification & Test Execution Evidence

Executed full test suite with 100% test passage (29/29 tests across 6 test suites):

```
> scopepilot-backend@1.0.0 test
> jest --detectOpenHandles --runInBand

PASS tests/ai.routes.test.js
PASS tests/validation.test.js
PASS tests/ai.mockCases.test.js
PASS tests/health.test.js
PASS tests/notFound.test.js
PASS tests/ai.contracts.test.js

Test Suites: 6 passed, 6 total
Tests:       29 passed, 29 total
Snapshots:   0 total
Time:        1.764 s
Ran all test suites.
```

8. Questions Prepared for Mentor Evaluation

Mentor Question	Prepared Technical Response
Why not use an LLM immediately?	A rule-based mock helps the team confirm the contract, validate edge cases, and establish predictable baseline behavior without adding model latency, costs, or non-deterministic variance.
What is partial overlap?	Part of the request is covered by existing deliverables (e.g. Catalog UI), while other parts introduce substantial new work (e.g. bulk CSV processing or background sync) that require extra pricing and scheduling.
What does the mock prove?	It proves the end-to-end data pipeline, Zod schema validation, case handling across 5 distinct outcomes, and client response structure before semantic AI is added.

9. Next Steps for Review 3

1. Introduce text extraction and semantic chunking for approved proposal PDFs/TXTs in `backend/src/ai/rag/`.
2. Wrap local embedding models (`nomic-embed-text`, 768 dimensions) under `backend/src/ai/embeddings/`.
3. Compare natural-language requests against approved-scope passages using vector similarity search, recording both successful semantic matches and retrieval failure cases.